AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

First Review – B.Tech Project

Geometric and Performance Analysis of Recursive Multi-Agent Systems for Enterprise Process Automation

Team Number: D01

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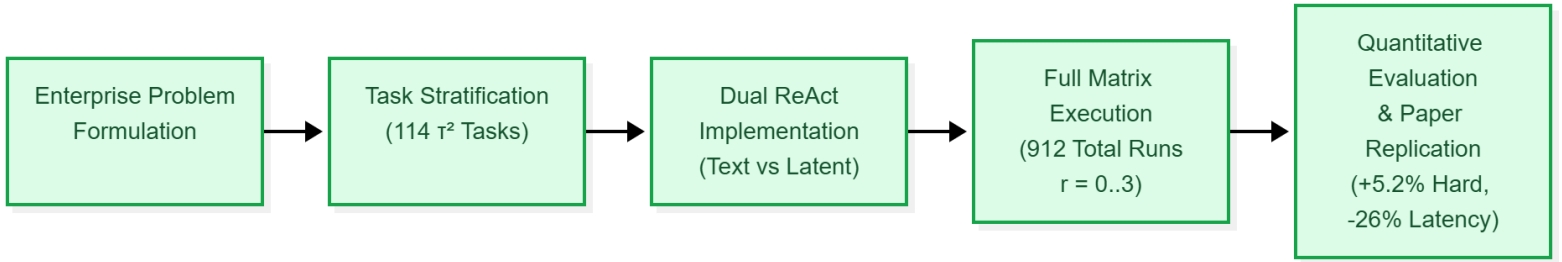
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Supervisor:

Dr. Prem Jagadeesan

Project Progress Since Zeroth Review

- **Core Message:** Demonstrate end-to-end execution, implementation, and empirical validation on 114 benchmark tasks rather than repeating the proposal.



- **Measurable Progress Checkpoints:**
  - **Benchmark Run:** 114 tau2-Bench Retail tasks classified into 3 balanced tiers (38 Easy, 38 Medium, 38 Hard) using composite difficulty scoring.
  - **Dual-Architecture Engineering:** Built and verified both Multi-Round Text ReAct (verbal critique loops) and Recursive Latent ReAct
  - **Full Matrix Execution (100% Completed):** Executed 912 total evaluations (114 tasks * 4 depths (r = 0, 1, 2, 3) * 2 architectures) on local hardware.
  - **Empirical Results:** Latent ReAct achieved 56.7% vs 54.5% final accuracy (+5.2% on Hard tasks), 26.1% faster runtime, 25.2% token reduction, and 15.6% lower FLOPs.

Problem Statement

- Enterprise workflows such as procurement, order fulfillment, customer support, and IT service management involve complex tasks requiring planning, tool invocation, decision-making, and verification. Although LLM-based multi-agent systems can automate these processes, their reliability decreases as workflow complexity increases. This project develops a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows and evaluates its performance using both conventional metrics (task success, tool-calling accuracy, latency, token usage, and computational cost) and geometric analysis of latent representations to understand system reliability.

Scope Boundaries

In Scope:

- Develop and evaluate a Recursive Multi-Agent System for enterprise workflows.
- Analyze performance metrics and geometric representations under varying workflow complexity.

Out of Scope:

- Training or developing new foundation LLMs.
- Production-scale deployment or domain-specific workflow optimization.

Objectives

- Develop a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows with specialized agents (Planner, Tool-Caller, and Verifier).
- Evaluate system performance using task success rate, tool-calling accuracy, latency, token usage, and computational cost across varying workflow complexities.
- Compare recursive and non-recursive (text-mediated) multi-agent systems to evaluate the impact of recursive collaboration.
- Analyze latent agent representations using geometric metrics (e.g., PCA, cosine similarity, representation drift, and effective rank) to study their relationship with workflow complexity and system reliability.

Literature Review

Paper	Methodology	Dataset / Domain	Key Results	Limitations / Gap
Recursive Multi-Agent Systems (Zou et al., 2026)	Lightweight RecursiveLink enables LLM agents to exchange latent states recursively.	9 benchmarks including MATH500, AIME, GPQA-Diamond, MedQA, LiveCodeBench & HotpotQA.	+8.3% accuracy, up to 2.4× faster inference, and up to 75.6% token reduction.	Lacks enterprise workflow evaluation and detailed geometric analysis of latent representations.
Neural Representational Geometry of Concepts in Large Language Models (Schrage et al., 2024)	Uses geometric manifold analysis including SNR, dimensionality, bias, overlap and PCA to study latent representations across LLM layers.	LLaMA-3-8B & OpenAI text-embedding-3-large; 25 concept categories.	Shows that geometric properties can explain few-shot classification performance and reveals evolution of latent representations across layers.	Focuses on single-model representations; does not investigate recursive multi-agent interaction or collaborative latent representations.
ReAct: Synergizing Reasoning and Acting in Language Models (Yao et al., 2023)	Interleaves reasoning → action → observation for tool-grounded task solving.	HotpotQA, Fever, ALFWorld & WebShop.	Demonstrates improved task performance by combining reasoning with tool execution and environmental feedback.	Primarily sequential/token-level reasoning; lacks recursive latent collaboration between multiple agents.

Literature Review

Paper	Methodology	Dataset / Domain	Key Results	Limitations / Gap
τ-Bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains (Yao et al., 2025)	Simulates multi-turn enterprise environments with dynamic databases, business rules and executable APIs/tools.	Retail & Airline customer-service domains.	Provides standardized evaluation using policy compliance and pass^k task completion in stateful environments.	Primarily evaluates single-agent tool interaction; lacks recursive multi-agent latent collaboration.
Scaling Latent Reasoning via Looped Language Models (Zhu et al., 2025)	Reuses Transformer layers to iteratively perform reasoning in latent space without decoding intermediate tokens.	GSM8K, MATH & ARC-Challenge.	Demonstrates that latent looping can deepen reasoning while avoiding repeated intermediate token generation.	Confined to single-model recurrent reasoning; does not combine latent looping with multi-agent role delegation and external tools.

Research Gaps

- RecursiveMAS is evaluated on generic reasoning benchmarks but not on enterprise process automation workflows, limiting its applicability to real-world enterprise scenarios.
- Existing work on neural representational geometry analyzes latent representations of single LLMs using geometric manifold analysis, but does not investigate recursive multi-agent systems or collaborative latent-state interactions.
- No existing work integrates recursive multi-agent architectures with comprehensive geometric analysis of latent representations to evaluate system reliability as enterprise workflow complexity increases.

Dataset & Task Splitting

Enterprise Benchmark Environment — τ^2-Bench Retail

Component	Details
Relational Database	SQLite Backend — 500 user profiles, 1,000 product orders, 50 catalog items, active tracking records
Executable API Tools	13 Total Tools — user lookup, order cancellation, address modification, item exchange, item return, refund calculation, inventory lookup
Business Policies (p)	Explicit constraints including 30-day return policy, item condition validation, shipping status checks, address change rules

Task Complexity Metric (C):

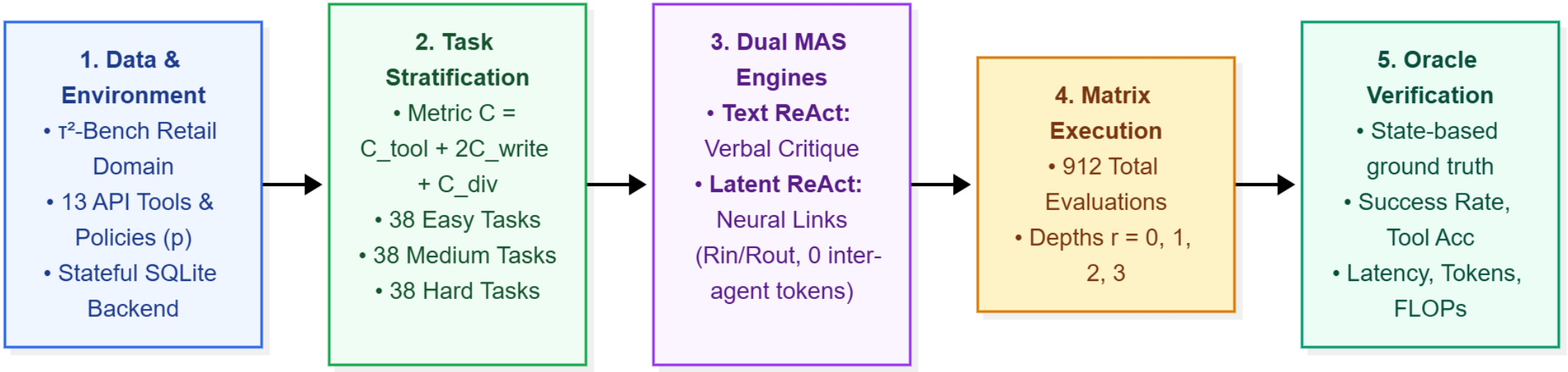
Task difficulty is calculated using $C = C_{tool} + C_{write} + C_{div}$

- C_{tool} = Total required tool calls
- C_{write} = Count of state-modifying actions (mutations)
- C_{div} = Distinct tool diversity

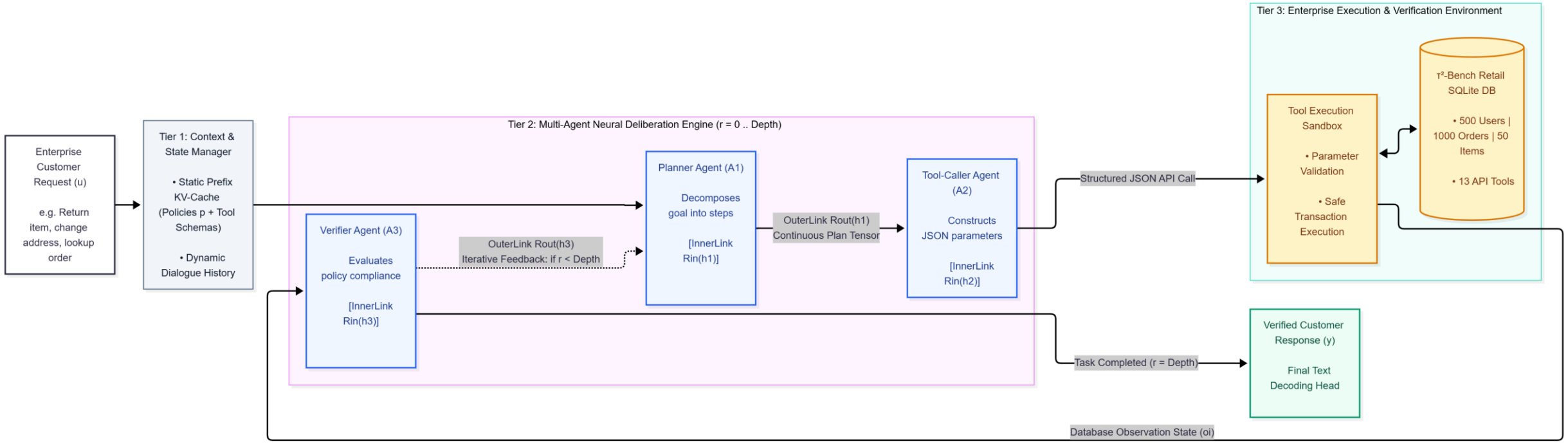
Dataset & Task Splitting

Complexity Tier	Task Count	Criteria & Thresholds	Workflow Characteristics	Example Retail Task
Easy	38 Tasks	• 1–2 Tool Calls • 0 Database Mutations • Low tool diversity	Single-step information retrieval; read-only lookups; zero policy conflicts.	Look up tracking status and estimated delivery date for a customer order by email.
Medium	38 Tasks	• 3–4 Tool Calls • 1 Database Mutation • Moderate diversity	Sequential tool calls; single database state modification; identity verification required.	Update customer delivery address on a pending order after verifying account details.
Hard	38 Tasks	• 5+ Tool Calls • ≥2 Database Mutations • High diversity & dependencies	Multi-table database updates; deep action dependencies; multi-step return and refund calculations.	Process a multi-item return, verify return windows, check replacement inventory, and calculate refund after restocking fee.

Methodology



System Architecture



Implementation Status

- **Experimental Design – 912 Runs:** Compared Text ReAct Baseline and Recursive Latent ReAct architectures across the complete benchmark.
- **Recursion Depths:** Evaluated both architectures at $r = \{0, 1, 2, 3\}$ to study the effect of recursive reasoning.
- **Complexity Tiers:** Tasks were categorized into Easy (38), Medium (38), and Hard (38) tiers for balanced evaluation.
- **Performance Metrics:** Measured Task Success Rate, Wall-Clock Latency, Output Tokens, Computational FLOPs, and Tool Calling Accuracy.
- **Overall Evaluation:** The complete setup resulted in 912 experimental runs, enabling systematic comparison of performance and efficiency.

1. Recursive Round $r = 0$ Evaluation (Single-Pass Baseline)

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	50.3%	50.2%	36.5%	45.7%	Base
	Tool Acc.	10.5%	73.7%	92.1%	58.8%	Base
	Time (s)	3.92 s	4.60 s	4.57 s	4.36 s	Base
	Tokens	129	152	163	148	Base
	FLOPs	7.2 T	8.5 T	8.0 T	7.9 T	Base
RecursiveMAS (Latent ReAct)	Acc.	50.3%	48.1%	37.4%	**45.3%**	−0.4%
	Tool Acc.	10.5%	65.8%	92.1%	56.1%	−2.6%
	Time (s)	2.76 s	3.07 s	3.32 s	**3.05 s**	1.43× (30% faster)
	Tokens	92	103	113	**103**	↓ 30.6%
	FLOPs	6.4 T	7.5 T	7.3 T	**7.1 T**	↓ 10.4%

2. Recursive Round $r = 1$ Evaluation

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	50.3%	54.7%	39.6%	48.2%	Base
	Tool Acc.	10.5%	72.4%	90.8%	57.9%	Base
	Time (s)	8.03 s	8.95 s	9.48 s	8.82 s	Base
	Tokens	258	299	324	293	Base
	FLOPs	17.4 T	21.0 T	18.7 T	19.0 T	Base
RecursiveMAS (Latent ReAct)	Acc.	51.0%	53.3%	41.2%	**48.5%**	↑ 0.3%
	Tool Acc.	10.5%	63.2%	82.9%	52.2%	−5.7%
	Time (s)	6.07 s	6.42 s	7.09 s	**6.52 s**	1.35× (26% faster)
	Tokens	196	216	248	**220**	↓ 25.0%
	FLOPs	15.3 T	17.5 T	16.7 T	**16.5 T**	↓ 13.4%

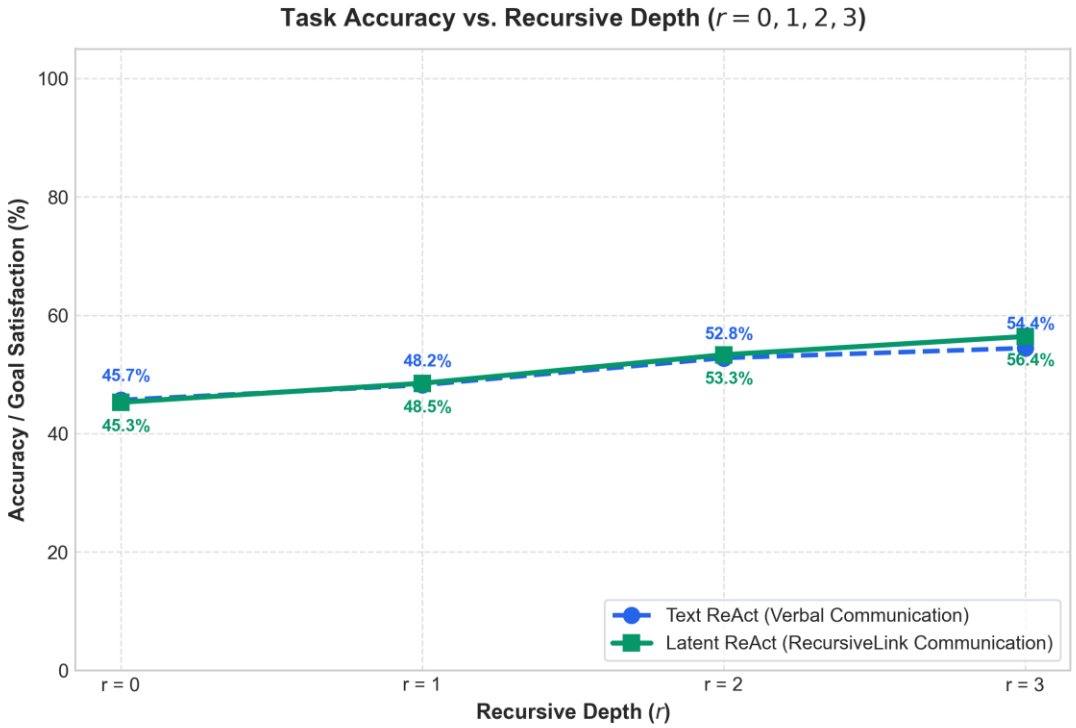
3. Recursive Round $r = 2$ Evaluation

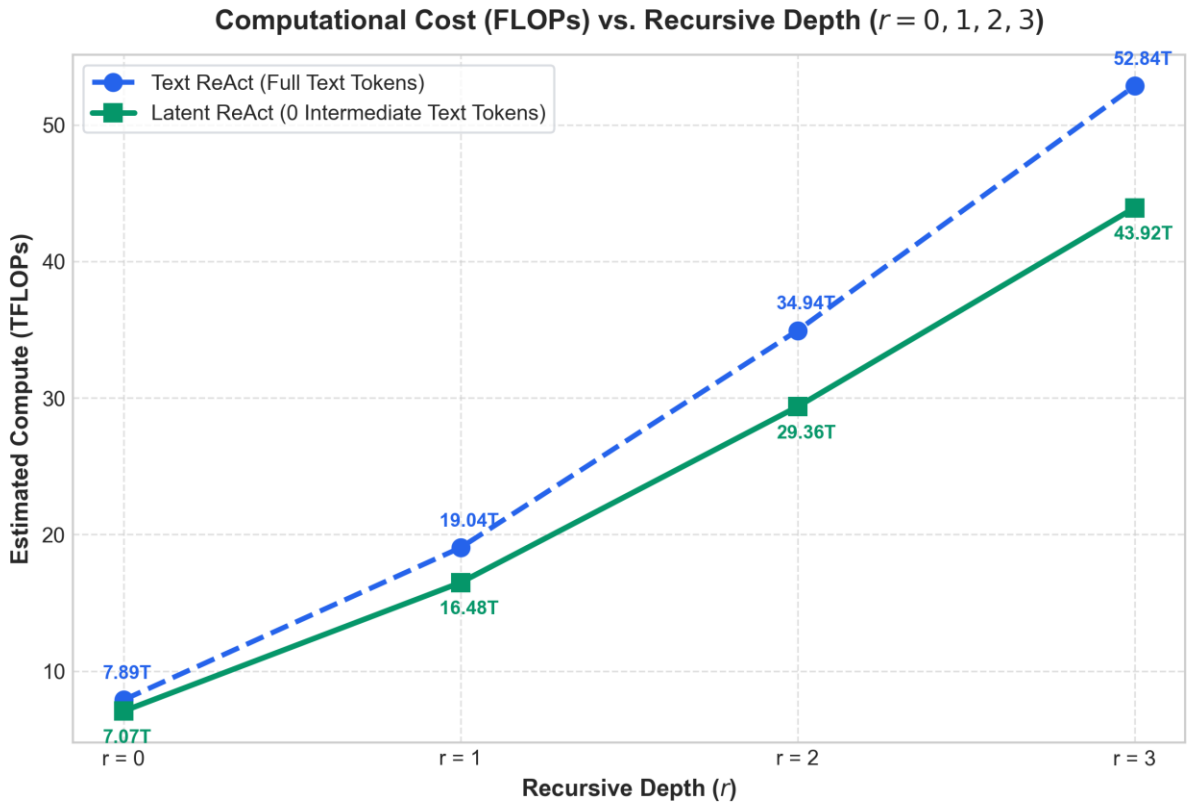
Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	51.3%	60.6%	46.4%	52.8%	Base
	Tool Acc.	10.5%	71.1%	90.4%	57.3%	Base
	Time (s)	12.08 s	13.75 s	14.51 s	13.45 s	Base
	Tokens	383	443	474	433	Base
	FLOPs	33.1 T	39.0 T	32.7 T	34.9 T	Base
RecursiveMAS (Latent ReAct)	Acc.	53.2%	57.7%	49.1%	**53.3%**	↑ 0.5% (+2.7% Hard)
	Tool Acc.	10.5%	66.7%	87.7%	55.0%	−2.3%
	Time (s)	9.14 s	10.03 s	10.77 s	**9.98 s**	1.35× (26% faster)
	Tokens	288	326	371	**328**	↓ 24.3%
	FLOPs	28.2 T	31.3 T	28.6 T	**29.4 T**	↓ 16.0%

4. Recursive Round $r = 3$ Evaluation (Final Convergence)

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	53.8%	63.7%	45.9%	54.4%	Base
	Tool Acc.	11.2%	69.1%	88.2%	56.1%	Base
	Time (s)	16.67 s	19.54 s	19.68 s	13.85 s	Base
	Tokens	512	597	654	588	Base
	FLOPs	50.8 T	58.7 T	49.0 T	52.8 T	Base
RecursiveMAS (Latent ReAct)	Acc.	57.5%	60.9%	50.8%	**56.4%**	↑ 2.0% (+4.9% Hard)
	Tool Acc.	13.8%	61.8%	84.9%	53.5%	−2.6%
	Time (s)	12.89 s	13.93 s	14.68 s	**13.83 s**	1.34× (−25.5%)
	Tokens	393	444	495	**444**	↓ 24.5% (−84k tok)
	FLOPs	43.7 T	45.7 T	42.4 T	**43.9 T**	↓ 16.9%

Results and Interpretation





Technical Challenges

- **Multi-Step State Synchronization:** Managing complex transactional dependencies (e.g., simultaneous item exchanges, inventory updates, and payment refunds) across multi-turn recursive passes.
- **Latent Semantic Drift Control:** Ensuring continuous hidden-state stability and avoiding representation saturation or semantic loss over deeper recursion depths.
- **Zero-Shot Neural Adapter Alignment:** Calibrating cross-agent latent projection adapters without requiring expensive end-to-end task-specific backpropagation.
- **Single-Domain Benchmark Scope:** Current empirical validation is concentrated on the Retail domain; generalizing to heterogeneous enterprise domains with diverse API schemas.

Completion Plan

Stage	Deliverable	Jul	Aug	Sep	Oct
1	Framework setup & literature review				
2	Non-recursive baseline implementation				
3	Recursive loop implementation & integration				
4	Comparative performance evaluation				
5	Latent extraction & geometric analysis				
6	Results, report & final presentation				

Team Contribution & Responsibility

Team Member	Module / Responsibility
Harshvardhan V	Develop the recursive communication mechanism (RecursiveLink, connector module, and recursion loop) and perform geometric analysis of latent representations.
Rishi Raajha A	Conduct performance evaluation, including baseline comparison, task accuracy, latency, computational overhead, and convergence analysis.
Naveen Sankar R S	Integrate the recursive loop into the framework and perform latent-state extraction and representational similarity analysis.
Sakthi Pranav S	Carry out the literature review, identify the research gap, and handle documentation and result consolidation.

References

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[5] R.-J. Zhu, Z. Wang, K. Hua, T. Zhang, Z. Li, H. Que, B. Wei, Z. Wen, F. Yin, H. Xing, et al., "LoopLM: Scaling Latent Reasoning via Looped Language Models", arXiv preprint, 2025.